

and an optional NEON media and floating-point processing engine. It is also scalable up to four coherent cores with advanced MPCore technology.

ARM's model of licensing its core processors to other vendors is a unique selling proposition. "Because desktop processors are really only available from two companies, there is limited innovation and diversity. Compare this to the ARM business model where any company can license an ARM processor and build its own unique intellectual property (IP) around its own unique configuration of ARM cores. This business model is very exciting as it provides opportunities for new companies in India to develop their own application processors for tablets or smartphones," says Bruce.

Intel Atom processors are also emerging and gaining a market in mobile devices. These are derivatives of desktop processors like Pentium Core i5, i7, etc. The Intel Atom processor Z6xx combines 45nm Intel Atom processor core with 3D graphics, video encode and decode, as well as memory and display controllers into a single SoC design. Combined with the Intel SM35 Express chipset, it supports Windows, MeeGo and Android operating systems.

While the processor appears more popular in the netbook space, a variety of smartphones and mobile Internet devices (MIDs) for cloud computing are also sporting the 'Intel Inside' logo these days. Joshi suggests that although ARM is a leader in the mobile space, Atom can benefit from the Intel's experience in the desktop space as the current trend portrays a deep convergence of desktop and mobile worlds.

## Multi-core magic

As mobiles race to beat the capabilities of computers, it is inevitable that some of the trends in the PC processor space will also spill over to the mobile world. One such trend is the magic of multi-cores.

"Due to the growth in the availability of high-speed mobile and WiFi

networks, mobile devices will also be used for various performance-intensive tasks that were previously handled by traditional PCs. Single-core mobile processors are not designed to deal with this tidal wave of high-performance use cases," explains Dhupar.

"On a mobile processor that has a multi-core CPU, multi-tasking can be shared between the distinct processing cores. Hence, as the performance requirements of mobile applications increase, SoC vendors are adopting multi-core processor architectures to deliver the increased performance and keep power consumption within mobile budgets," he adds.

It is a common misconception that more cores mean higher power consumption. In fact, multi-core CPUs are able to distribute their workload across their cores so that each CPU core can run at a lower frequency and voltage. This means each core consumes significantly lower power and offers much higher performance per watt than single-core CPUs.

The performance of a smart phone has increased 40 times over the last ten years; and the increase in performance since 2008 alone has been eight times. One reason for this is the adoption of multi-core technology. Last year saw the launch of dual-core Cortex-A9 smartphones. This year, we are likely to see the launch of quad-core Cortex-A9 smartphones.

Bruce predicts that at the end of 2012, we will see the launch of Cortex-A15 handsets with a 50 per cent increase in performance. "Also in 2012 you will see the launch of Cortex-A15 based handsets at a \$100 price point. The Cortex-A15 processor allows \$100 smartphones to deliver the user experience of a \$500 smartphone in 2010," he says.

NVIDIA is also focusing on multi-core architectures. Its latest Tegra 3 processor goes a step beyond quad-core by adding a fifth 'companion' core. Its internal architecture is identical to the four main Cortex-A9 CPU cores but the fifth core is built using a special low-power silicon process. Using a technology called variable

symmetric multiprocessing, the fifth core handles low-frequency tasks such as those common in active standby mode (Twitter and Facebook updates, e-mail synchronisation, etc) and applications that do not require significant CPU processing power, such as audio streaming, offline audio playback, and both online and offline video playback.

When more demanding tasks are required, the other four cores can be called upon. This approach allows Tegra 3 to deliver significantly lower power than competing mobile processors at all performance levels.

It is interesting to note that the popular low-cost, open mobile software development platform PandaBoard too is now available in a dual-core version, keeping pace with the current trend. In November 2011, the community announced the availability of the PandaBoard ES based on TI's OMAP4460 processor, whose multi-core architecture includes two ARM Cortex-A9 cores running at up to 1.2 GHz each, delivering a 20 per cent increase in overall performance and a 25 per cent increase in graphics when compared to the OMAP4430 processor used by the earlier PandaBoard.

## Looking ahead

Dhupar feels that with mobility becoming more sophisticated, the number of CPU cores in a mobile processor will continue to multiply. Quad-core CPUs will enable mobile devices to further push the performance envelope and allow application and game developers to deliver new mobile experiences, all while extending the battery life.

Dhupar expects Windows on ARM platform becoming a reality some time late this year or next year.

Joshi sums up, "The rapid increase in consumer demand for multimedia, gaming, etc on mobile devices will drive computing speed. Aside from that, low power consumptions, size and weight are key improvements to foresee." ●

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